

平成18年度 筑波大学大学院 博士前期課程 システム情報工学研究科
コンピュータサイエンス専攻 入学試験問題

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基礎課題 (数学と情報基礎の全ての問題に解答すること)

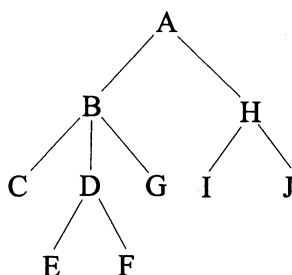
じょうほう きそ
情報基礎

以下の問題 I、問題 II を解答せよ。各解答は別々の答案用紙に記入すること。

Solve both Problem I and Problem II, answering in two different sheets for them.

Problem I

- (1) Consider the following ordered tree rooted by node A . Suppose to traverse this tree in preorder and assign integers $1, 2, 3, \dots$ to each node based on the visiting order. Let us denote the value for node v as a function $\text{pre}(v)$. Then we get $\text{pre}(A) = 1$, $\text{pre}(B) = 2$, $\text{pre}(C) = 3$, $\text{pre}(D) = 4$, $\text{pre}(E) = 5$, $\text{pre}(F) = 6$, $\text{pre}(G) = 7$, $\text{pre}(H) = 8$, $\text{pre}(I) = 9$, and $\text{pre}(J) = 10$.



Similarly, consider to traverse the tree in postorder and assign integers $1, 2, 3, \dots$ to each node based on the visiting order. Let us denote this function as $\text{post}(v)$. Show the values of $\text{post}(A)$, $\text{post}(B)$, $\dots$, $\text{post}(J)$.

- (2) We can use functions $\text{pre}()$ and $\text{post}()$ defined in (1) to decide whether two nodes in an ordered tree have an ancestor-descendant relationship. Show the necessary and sufficient condition for node v to become the ancestor of node v' by using $\text{pre}()$ and $\text{post}()$ and explain why the condition holds.

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(Continued from the previous page)

(3) We can represent the tree given in the previous page as a string:

$\langle A \rangle \langle B \rangle \langle C \rangle \langle /C \rangle \langle D \rangle \langle E \rangle \langle /E \rangle \langle F \rangle \langle /F \rangle \langle /D \rangle \langle G \rangle \langle /G \rangle \langle B \rangle \langle H \rangle \langle I \rangle \langle /I \rangle \langle J \rangle \langle /J \rangle \langle H \rangle \langle /A \rangle$

where $\langle A \rangle$ and $\langle /A \rangle$ are called an open tag and an end tag for A , respectively, and A is called the label for $\langle A \rangle$ and $\langle /A \rangle$.

Arbitrary combinations of open and end tags may make illegal strings such as $\langle A \rangle \langle /B \rangle \langle C \rangle$ and $\langle A \rangle \langle B \rangle \langle C \rangle \langle /B \rangle \langle /C \rangle \langle /A \rangle$; they do not correspond to any tree structures. Consider to construct a function `check_tags( )` using a stack. If a given string corresponds to a tree structure, the function returns *true* and otherwise it returns *false*. Write pseudo-codes to be included in (a) and (b) in the following algorithm to accomplish the purpose. You can write pseudo-codes using a natural language.

function `check_tags( )`

while more tags **do**

 read next *tag*

if *tag* is an open tag

 push *tag* to the stack

else /* *tag* is an end tag */

(a)

end

end

(b)

end

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Problem II

The Taylor expansion of $\sin x$ is given by

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Answer the following questions.

- (1) Write the function `double factorial(int n)` returns the factorial of a positive integer n .
- (2) The following function `double power(double x, int n)` returns x^n , where x is a real number and n is a positive integer, respectively. For example, if $n = 11$ is given to the function, the values of n , y and z after line 11 are as follows:

n	y	z
5	x	x^2
2	x^3	x^4
1	x^3	x^8
0	x^{11}	x^{16}

Complete the following definition of the function `power` by filling the boxes (a) and (b).

```

1  double power(double x, int n)
2  {
3      double y, z;
4      y = 1.0;
5      z = x;
6      while (n > 0) {
7          if ( (a) ) {
8              y *= z;
9          }
10         (b)
11         n /= 2;
12     }
13     return y;
14 }
```

- (3) Write the function `double my_sin(double x, int n)`, which returns the Taylor expansion of $\sin x$ up to n terms, by using the above functions `double factorial(int n)` and `double power(double x, int n)`.